

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheets for Field Cold Bends
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-007
Document Revision	R01
Document Status	Issued For Review
Originator	Sunday Ikechukwu
Security Classification	Restricted
Issue Date	10 Aug 2022

Data Sheets for Field Cold Bends

R01	10.08.22	Issued For Review	S.Ikechukwu	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	10.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-007

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

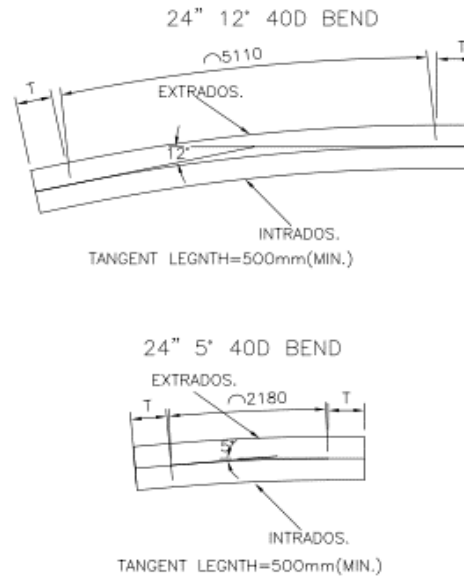
REFERENCES

Document No.	Description

Table of Contents

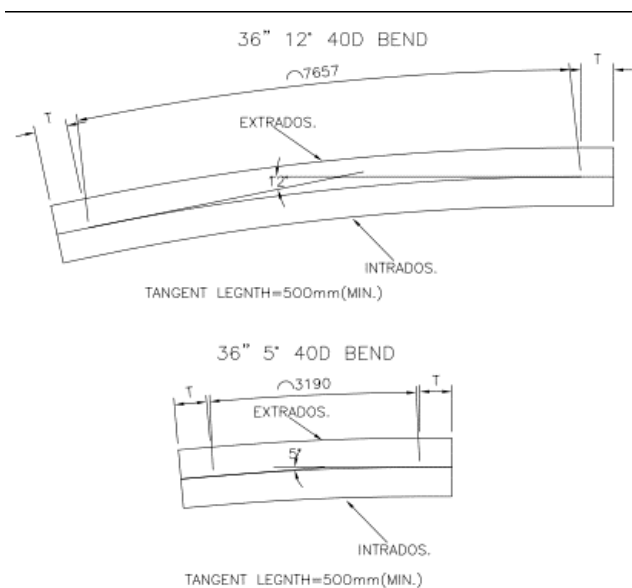
1. DATA SHEET FOR COLD BEND- 24" ND	5
2. DATA SHEET FOR COLD BEND- 36" ND	7

1	DATA SHEET FOR COLD BEND- 24" ND		
2	Bend Characteristics		
3	Fluid	Multiphase (Gas and Condensate)	
4	Mother Pipe Material	API 5L GR. X70	
5	Mother Pipe NPS	24"	
6	Manufacturing Process	Cold Bending	
7	Minimum Bend Radius	40*D, Tolerance of ±1.5 mm	
8	Pressure Rating	900#	
9	Bend Angle	1° to 12° (Allowed)	
10	Min.Tangent Lengths (mm)	500	
11	Chemical composition	Shall comply with mother pipe requirement	
12	Mother Pipe Diameter Outside Diameter (mm)	610 mm	
13	Mother Pipe Wall Thickness (mm)	15.88mm	
14	Applicable Design Code	ASME B31.8 and ISO 15590-1.	
15	Design Conditions		
16	Design Pressure (Barg)	110.00	
17	Design temperature (°C)	60.00	
18	Corrosion Allowance (mm)	3mm	
19	Hydrotest Presure (Barg)	137.5	
20	Other Requirements		
21	Bend End Finish	Bevel Ends	
22	External Coating	3LPE (same as Mother Pipe)	
23	Dimensions		
24	Out-of-roundness	1% for bend end and 2.5% for bend body	
25	Out-of-planeness	1 % Of OD As per DEP 31.40.20.33-Gen	
26	End out-of-squareness	Tolerance of ±3mm	
27	Bevel End	30° with tolerance as provided for the pipe	
28	Testing & Inspection / Acceptance Requirements		
29	Qualification	As per DEP 31.40.20.37-Gen	
30	Ultrasonic	Ultrasonic test shall be performed on the outer surface of bends if required	
31	Magnetic Particle Inspection	N/A	
32	Hardness (maximum)	N/A	
33	Residual magnetism	As per DEP 31.40.20.37-Gen	
34	Marking		
35	Marking (location)	N/A	
36	Marking	N/A	
37	Standard Marking	N/A	
38	Bend Length Calculation		
39	Bend Length, L _b (Formula)		((Bend Radius x Bend Angle x pi / 180)) + (2xtangent length)) / 500)
40	Bend Angles (variable)	Degree	As determined by Site Conditions.
41	Bend Length (including tangent length)	m	To be calculated by Contractor.
42	Quantities (See Note 3)	Pcs	As determined by Conditions.
43	Note		
44	1. The line pipe to be used for the Field / Cold Bending must be the same wall thickness with the adjacent straight run pipes on both sides of the bends. Installation of the bends inbetween adjacent runs of line pipes with differen wall thicknesses shall be discourraged.		
45	2. The wall thickness of the bends due to stretching shall not be less 95% of the starting wall thickness or less than the nominal wall thickness minus the manufacturing tolerance which ever one that is greater.		
46	3. Ovality of pipe cross section after bending shall not exceed 2.5 percent		
47	4. Each field Bend shall be checked for coating damaging or disbonding from the pipe for repairs.		
48	3. Quantities shall be comfirmed after Detailed Topographical Survey		

Fig 1: Typical Bend Detail FOR 24" BEND FOR 12° AND 5°**Table 1: REFERENCES:**

REF.	DOCUMENT NUMBER	TITLE
1	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-007	Specification for Cold Bend
3	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipeline Wall Thickness Calculations
4	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-001	Data Sheet for Line Pipe
10	ASME B31.8	Gas Transmission and Distribution Piping Systems

1	DATA SHEET FOR COLD BEND- 36" ND		
2	Bend Characteristics		
3	Fluid	Multiphase (Gas and Condensate)	
4	Mother Pipe Material	API 5L GR. X70	
5	Mother Pipe NPS	36"	
6	Manufacturing Process	Cold Bending	
7	Minimum Bend Radius	40*D, Tolerance of ±1.5 mm	
8	Pressure Rating	900#	
9	Bend Angle	1° to 12° (Allowed)	
10	Min.Tangent Lengths (mm)	500	
11	Chemical composition	Shall comply with mother pipe requirement	
13	Mother Pipe Diameter Outside Diameter (mm)	914 mm	
14	Mother Pipe Wall Thickness (mm)	23.83mm	
15	Applicable Design Code	ASME B31.8 and ISO 15590-1.	
16	Design Conditions		
17	Design Pressure (Barg)	110.00	
18	Design temperature (°C)	60.00	
19	Corrosion Allowance (mm)	6 mm	
20	Hydrotest Presure (Barg)	137.5	
21	Other Requirements		
22	Bend End Finish	Bevel Ends	
23	External Coating	3LPE (same as Mother Pipe)	
24	Dimensions		
25	Out-of-roundness	1% for bend end and 2.5% for bend body	
26	Out-of-planeness	1 % Of OD As per DEP 31.40.20.33-Gen	
27	End out-of-squareness	Tolerance of ±3mm	
28	Bevel End	30° with tolerance as provided for the pipe	
29	Testing & Inspection / Acceptance Requirements		
30	Qualification	As per DEP 31.40.20.37-Gen	
31	Ultrasonic	Ultrasonic test shall be performed on the outer surface of bends if required	
32	Magnetic Particle Inspection	N/A	
33	Hardness (maximum)	N/A	
34	Residual magnetism	As per DEP 31.40.20.37-Gen	
35	Marking		
36	Marking (location)	N/A	
37	Marking	N/A	
38	Standard Marking	N/A	
39	Bend Length Calculation		
40	Bend Length, L _b (Formula)		((Bend Radius x Bend Angle x pi / 180)) + (2xtangent length) / 500)
41	Bend Angles (variable)	Degree	As determined by Site Conditions.
42	Bend Length (including tangent length)	m	To be calculated by Contractor.
43	Quantities (See Note 3)	Pcs	As determined by Conditions.
44	Note		
45	1. The line pipe to be used for the Field / Cold Bending must be the same wall thickness with the adjacent straight run pipes on both sides of the bends. Installation of the bends inbetween adjacent runs of line pipes with differen wall thicknesses shall be discouraged.		
46	2. The wall thickness of the bends due to stretching shall not be less 95% of the starting wall thickness or less than the nominal wall thickness minus the manufacturing tolerance which ever one that is greater.		
47	3. Ovality of pipe cross section after bending shall not exceed 2.5 percent		
48	4. Each field Bend shall be checked for coating damaging or disbonding from the pipe for repairs.		
49	3. Quantities shall be comfirmed after Detailed Topographical Survey		

Fig 1: Typical Bend Detail FOR 36" BEND FOR 12° AND 5°**Table 1: REFERENCES:**

REF.	DOCUMENT NUMBER	TITLE
1	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-007	Specification for Cold Bend
3	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipeline Wall Thickness Calculations
4	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-001	Data Sheet for Line Pipe
10	ASME B31.8	Gas Transmission and Distribution Piping Systems